PID MOTOR CONTROL LOOP

Jason Gallicchio and Claude, rev A, 2026-10-29

LAB 10L: a PID loop holds a fader (motorized potentiometer) where TARGET says.

POWER: +15 V, GND, −15 V on the POWER terminal, 1 A per rail.
    Both green lamps light. A reversed supply is blocked: the lamps stay dark.
FADER (Bourns PSM01-082A-103B2): motor A to MOTOR, motor B to GND.
    Track 1, 2, 3 to POT 1 (+), 2 (wiper), 3 (−): the numbers match.
    Check with LOOP open: slide it by hand, POSITION is +7.7 V at the + end.
    LOOP closed and the slider runs to an end and stays: swap the motor leads.
INPUTS
    TARGET knob: centre detent = 0 V, clockwise +, up to +/−6.4 V.
    TARGET IN jack: a function generator, +/−12 V max.
        Try +/−3 V: a 0.2 Hz square wave for steps, a 0.05 Hz triangle for stick-slip.
OUTPUTS, each through 1k: for a scope, not a load
    POSITION OUT: the fader's wiper, buffered. DRIVE OUT: the motor drive.
SWITCHES

TARGET SOURCE	1 = the knob
	3 = the TARGET IN jack
LOOP	1 = open: no feedback; DRIVE = P GAIN x TARGET (+ D, I if in)
	P GAIN anticlockwise, D and I off: DRIVE = TARGET
	3 = closed: POSITION fed back, the loop holds TARGET
D IN / OFF	1 = off: the differentiator's output shorted
	3 = D into the sum
D CAPACITOR	1 = 22 nF, for this fader
	3 = 100 nF, for the book's slower rotary pot
I IN / OFF	1 = off: the 1 uF shorted, so I restarts from zero
	3 = I into the sum

TRIMMERS: clockwise = more resistance: more P, more D, but LESS I.
START: every switch on 1. P and D GAIN anticlockwise, I GAIN clockwise.
    LOOP to 3: the slider follows the knob. Raise P GAIN until it rings.
    D IN, raise D GAIN: the ringing dies. Lower P; I IN: the error goes to 0.
WARNINGS
    A short to GND opens the fuse in about a second (red lamps); it resets.
    A stall may never open it: HOT copper reaches 90 C, so use the standoffs.
    Never short MOTOR or DRIVE to a rail. Power off to wire the fader.
    Scope ground clips go on the GND loops only, never on a signal or a rail.

https://github.com/gallicchio/motor-pid
circuit, scripts, schematic, this board

PID Motor Control Loop rev A, 2026-10-29: bottom, seen from below (mirrored), scale 1.49:1